KNECK OFFICIAL

Agriculture — Paper 1

443/1 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all questions in Sections A and B and ANY TWO essays in Section C. Section A 30 marks, B 20, C 30.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

All questions; 30 marks.

1. State three benefits of studying agriculture. [3 marks]

2. Name three physical components of soil. [3 marks]

3. Give three reasons for soil sampling before fertiliser use. [3 marks]

4. State three nursery-management practices. [3 marks]

5. Give three reasons for crop rotation. [3 marks]

6. State three signs of mature maize suitable for dry harvesting. [3 marks]

7. Give three methods of soil and water conservation. [3 marks]

8. Name three farm records and one use of each. [3 marks]

9. State three risks of poor pesticide handling. [3 marks]

10. Give three benefits of agroforestry. [3 marks]

SECTION B

All questions; 20 marks.

11. A field is 40m by 25m. Fertiliser rate is 200kg/ha. (a) Find area in hectares. (2) (b) Find fertiliser required. (2) (c) Explain three factors determining correct fertiliser choice. (6) [10 marks]

12. An irrigation scheme uses furrows. (a) Explain two advantages and two limitations. (4) (b) State three ways to improve water-use efficiency. (3) (c) Explain three signs of water stress in a crop. (3) [10 marks]

SECTION C

Choose two; 15 marks each.

13. Describe production of dry beans from land preparation to storage under local conditions. Include suitable variety/seed, planting, nutrient management, weed/pest control, harvest and safe storage. [15 marks]

14. Explain five causes of soil fertility decline and five sustainable responses; describe one method of monitoring improvement. [15 marks]

15. Describe how to establish and manage a vegetable nursery and transplant seedlings successfully. [15 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (3 marks) Food security,employment/income,resource management,skills,any three.

2. (3 marks) Mineral particles,organic matter,water,air,any three (organisms acceptable if described as living component).

3. (3 marks) Determine fertility/pH,target nutrients,avoid waste/pollution,guide amendments,three.

4. (3 marks) Watering,shade,mulching,weeding,thinning,pest control,hardening,three.

5. (3 marks) Break pests/disease,balance nutrients,weed control,improve soil,three.

6. (3 marks) Drying husks,hard grain,black layer/drying plant,three with correct crop context.

7. (3 marks) Terraces,contour farming,mulch,cover crops,grass strips,three.

8. (3 marks) Production→performance; financial→cost/profit; breeding/health→management,one paired mark each.

9. (3 marks) Poisoning,residues,water/soil contamination,non-target harm,resistance,three.

10. (3 marks) Soil protection,nutrient cycling,fodder/fuel/income,shade,biodiversity,three.

SECTION B

11. (10 marks) Area1000m^2=.1ha (2);20kg (2). Soil test/crop need,nutrient analysis,pH/soil conditions,cost/timing,three explained (6).

12. (10 marks) Simple/low energy if gravity; losses/waterlogging/erosion depending context, four distinct (4). Schedule by need,repair leaks,mulch/level/drip where suitable,three (3). Wilting,leaf rolling,reduced growth/scorching,three (3).

SECTION C

13. (15 marks) Land/site3; certified suitable seed/spacing3; soil-test nutrient plan2; early weeding/IPM3; harvest/dry/thresh2; clean dry pest-proof storage2. Accept locally appropriate agronomy, not one universal chemical rate.

14. (15 marks) Causes five distinct5; matched sustainable responses five5; monitoring plan soil test/organic matter/crop yield with baseline,repeat timing,consistent method,records and interpretation5.

15. (15 marks) Site/water/soil3; bed preparation/seed choice/sowing3; watering/shade/mulch3; sanitation/thinning/hardening3; correct transplant timing,root care/spacing/watering3.

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf